



Zhenyu Fan

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EDUCATION

- **Peking University** – Beijing, China Sep 2023 - June 2027 (expected)
PhD in Pure Mathematics
Advisors: Prof. Qing Han, Prof. Yuguang Shi and Prof. Yu Yuan
- **Xi'an Jiaotong University** – Xi'an, China Sep 2019 - June 2023
Bachelor, Honors Science Program in Mathematics
Advisor: Prof. Dongsheng Li
- **Xi'an Jiaotong University** – Xi'an, China Sep 2017 - June 2019
Young Gifted Class

RESEARCH INTEREST

- **Nonlinear Elliptic Equations & Geometric Analysis**
Especially minimal surfaces, Monge-Ampère equations, special Lagrangian equations and other fully nonlinear equations

RESEARCH PAPERS

- [8] **Zhenyu Fan**. Perturbation of special Lagrangian cones near infinity. *In preparation*. (2026)
- [7] **Zhenyu Fan**, Ravi Shankar. Alexandrov regularity for solutions to fully nonlinear curvature equations . *In preparation*. (2026)
- [6] **Zhenyu Fan**. Geometric regularity for semi-convex viscosity solutions to the special Lagrangian equation. *In preparation*. (2026)
- [5] **Zhenyu Fan**, Caiyan Li, Zhehui Wang. Failure of interior Hölder and gradient estimates for the subcritical special Lagrangian equation. *preprint, arXiv:2609.01531. Submitted to Calc. Var. Partial Differential Equations*. (2026)
- [4] **Zhenyu Fan**, Ravi Shankar. Interior estimates and regularity for the scalar curvature equation in dimension 4. *preprint, arXiv:2608.22748. Submitted to Geom. Func. Anal.* (2026)
- [3] **Zhenyu Fan**. Hessian estimates for the sigma-2 equation with variable right-hand side terms in dimension 4. *Advances in Mathematics* Volume 494 (2026)
- [2] **Zhenyu Fan**. A generalization of Savin's small perturbation theorem for fully nonlinear elliptic equations and applications. *Calculus of Variations and Partial Differential Equations* **65**, 182 (2026)
- [1] **Zhenyu Fan**, Yugao Ouyang. Asymptotic behavior of exterior minimal graphs in half spaces. *The Journal of Geometric Analysis* **35**, 247 (2025)

AWARDS & HONORS

- **Presidential Scholarship**, Peking University 2026
- **Zhou Yulin Scholarship**, Peking University 2025
- **Award for Scientific Research**, Peking University 2024
• **Pingfanghe Investment Scholarship**, Peking University
- **Outstanding Graduates**, Xi'an Jiaotong University 2023

INVITED TALKS

3. **Title:** On the sigma-2 equation and the scalar curvature equation in dimension 4.
Seminar, Academy of Mathematics and Systems Science, CAS, Beijing, China.

April 9, 2026
2. **Title:** On the sigma-2 equation and the scalar curvature equation in dimension 4.
Seminar, Xi'an Jiaotong University, Xi'an, China.

March 6, 2026
1. **Title:** On the sigma-2 equation and the scalar curvature equation in dimension 4.
Doctoral Forum on Elliptic PDEs, Fudan University, Shanghai, China.

December 3, 2025

TEACHING EXPERIENCE

Peking University, Teaching assistant

- Advanced Math A

Spring 2026
- Advanced Math A

Fall 2025
- Advanced Math B

Spring 2025
- Advanced Math B

Fall 2024
- Advanced Math B

Spring 2024
- Advanced Math C

Fall 2023

ACADEMIC SERVICES

Referee for several journals including: *Calculus of Variations and Partial Differential Equations*